GPS Data Analysis & Telemetry Report

Date of Analysis: March 13, 2026

Data Source: Serial Monitor Logs (Raw NMEA & Parsed Telemetry)

Location Derived: Surat, Gujarat, India (Lat: 21.160312, Lon: 72.788742)

1. Executive Summary

This report analyzes a sample of GPS data output containing both raw NMEA 0183 sentences and a parsed telemetry table. The GPS module is successfully achieving a highly accurate, stable 3D fix using multiple satellite constellations (GPS and BeiDou). The serial communication is perfectly stable with zero data corruption detected.

2. Raw NMEA Sentence Analysis

The GPS receiver outputs standard NMEA sentences. The prefix dictates the constellation: `$GP` (US GPS), `$BD` (China BeiDou), and `$GN` (Multi-GNSS combo).

Observed Sentences:

- `$GNGGA` **(Global Positioning System Fix Data):** The primary location sentence. Contains time, latitude, longitude, fix quality, number of satellites, HDOP, and altitude.
- `$GPGSA` / `$BDGSA` **(GNSS DOP and Active Satellites):** Lists the specific satellites used for the position fix and provides the Dilution of Precision (DOP) values.
- `$GPGSV` / `$BDGSV` **(GNSS Satellites in View):** Provides detailed information about all satellites the antenna can "see," including their ID (PRN), elevation, azimuth, and Signal-to-Noise Ratio (SNR).
- `$GNRMC` **(Recommended Minimum Specific GNSS Data):** Essential navigation data including position, velocity (speed), track/course, and date.
- `$GNZDA` **(Time & Date):** Provides high-precision UTC time, day, month, year, and local time zone offset.

Example Breakdown: `$GNGGA`

```
$GNGGA,190243.000,2109.61898,N,07247.32337,E,1,16,0.63,43.2,M,-59.8,M,,*65
```

Field	Value	Interpretation
Time (UTC)	`190243.000`	19:02:43 (7:02:43 PM)
Latitude	`2109.61898, N`	21 Degrees, 09.61898 Minutes North
Longitude	`07247.32337, E`	072 Degrees, 47.32337 Minutes East

Fix Quality	`1`	Valid Standard GPS Fix
Satellites Tracked	`16`	16 active satellites contributing to the fix
HDOP	`0.63`	Horizontal Dilution of Precision (Excellent)
Altitude	`43.2, M`	43.2 Meters above mean sea level

3. Parsed Telemetry Data Breakdown

The secondary data set shows the NMEA strings parsed into a structured format (likely via a microcontroller library such as TinyGPS++).

Positional & Fix Quality Metrics

- **Sats (Satellites):** `9` - The number of satellites currently used to compute the position. 9 guarantees a highly reliable 3D fix.
- **HDOP:** `0.9` - Horizontal Dilution of Precision. A value under 1.0 represents "ideal" positional geometry. The margin of error is exceptionally low.
- **Latitude (deg):** `21.160312` - Decimal format of the North coordinate.
- **Longitude (deg):** `72.788742` - Decimal format of the East coordinate.
- **Alt (m):** `~39.70` - Altitude in meters above mean sea level.

Timing & Freshness

- **Date:** `02/19/2026` - The UTC date of the recorded log.
- **Time:** `18:38:59` to `18:39:06` - The UTC timestamps of the parsed logs.
- **Fix Age / Date Age / Alt Age:** Ranging from `14ms` to `901ms`. This indicates how much time has passed since the library received an update for these specific parameters. Values well under 1000ms indicate a healthy ~1Hz to 10Hz update rate.

Kinematic Data

- **Speed:** `0.00` - The receiver is currently stationary.
- **Course from GPS:** `117.72` (Card: `ESE`) - The current directional heading calculated by the GPS module. Since the speed is 0, this value is residual or drifting and can be ignored until movement begins.
- **Distance / Course to London:** `7014 km` at `318.14 (NW)` - A calculated Haversine/Great Circle distance from the current coordinates to a hardcoded waypoint (London).

System Diagnostics

- **Chars RX:** Ramping up to `4527` - Total characters successfully read over the UART/Serial bus.

- **Sentences RX:** Ramping up to `16` - Total valid NMEA sentences successfully parsed.
- **Checksum Fail:** `0` - **CRITICAL METRIC.** A value of 0 indicates flawless electrical communication between the GPS module and the host processor. There is no electromagnetic interference (EMI) or baud rate mismatch corrupting the serial lines.

4. Conclusion & System Health

The GPS subsystem is performing optimally.

1. **Signal Integrity:** The module has a clear view of the sky, tracking an excellent number of satellites (16 visible in raw, 9 used in parsed) yielding a superb HDOP of 0.9.
2. **Hardware Health:** Perfect UART communication is confirmed by the 0 checksum failures over 4500+ characters received.
3. **Readiness:** The data is clean, parsed correctly, and ready to be used in navigation algorithms, Kalman filters, or data-logging applications.